Tutorial 6
CSE 112 Computer Organization

Q1.

```
n=input()

for(int i=n;i!=0;i--){
      for(int j=i;j!=0;j--)”{
            cout<<” * ”<<” ” <<;
      }
}
cout<<endl;
```

--

```
in R1                                     // Used as loop 1 iterator
mov R15, #1
loop1:
      beq R1, #0, loop1_exit              // loop1’s exit condition
      mov R2, R1                          // R2 is loop2’s iterator
      loop2:
            beq R2, #0, loop2_exit
            out “* “
            sub R2, R2, R15
            beq R15, #1, loop2            // Unconditional Jump
      loop2_exit:
      out “\n”                            // Print new line
      sub R1, R1, R15                     // loop1’s iterator update
      beq R15, #1, loop1                  // Unconditional Jump
loop1_exit:
hlt
```

Q2:

```
input n
switch(n):
     case 5: print 5;
     case 4: print 4;
     case 3: print 3;
     case 2: print 2;
     case 1: print 1;
     case 0: print 0;
```

```
in R0
beq R0, #5, case5
beq R0, #4, case4
beq R0, #3, case3
beq R0, #2, case2
beq R0, #1, case1
beq R0, #0, case0
case 5: out "5 "
case 4: out "4 "
case 3: out "3 "
case 2: out "2 "
case 1: out "1 "
case 0: out "0 "
hlt
```

Q3

```
n = input()
switch(n) {
     case 0: print 0;
             break;
     case 1: print 1;
             break;
     case 2: print 2;
             break;
     case 3: print 3;
             break;
     case 4: print 4;
             break;
     case 5: print 5;
             break;
     default : print Invalid;
               break;
     }
```

```
in R0
mov R1, #1
beq R0, #5, case5
beq R0, #4, case4
beq R0, #3, case3
beq R0, #2, case2
beq R0, #1, case1
beq R0, #0, case0
out “Invalid”
beq R1, #1, exit
case 5: out “5“
        beq R1, #1, exit
case 4: out “4“
        beq R1, #1, exit
case 3: out “3“
        beq R1, #1, exit
case 2: out “2“
        beq R1, #1, exit
case 1: out “1“
```

```
        beq R1, #1, exit
case 0: out “0“
        beq R1, #1, exit
exit: hlt
```
